

Creator Defense

Project: Verified Study Buddy

1. Your mission

Your learning target

I can explain the mechanism and terminology of generative systems defense.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Capstone Evaluate and Improve. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Capstone Design. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Bias and Representation. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 36: Creator Defense

A creator explains architecture, evidence, limitations, provenance, and responsible use.



CREATOR MISSION

Demo and defend against fresh test and design questions.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall Capstone evaluation and revision. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

This week's end goal

By the end, I can explain creator defense using a traced example and a stated limitation.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use creator defense to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A generative-system defense presents the question, architecture, corpus, permissions, baseline, evaluation and unresolved failures. Demonstration fluency is only one observable property.

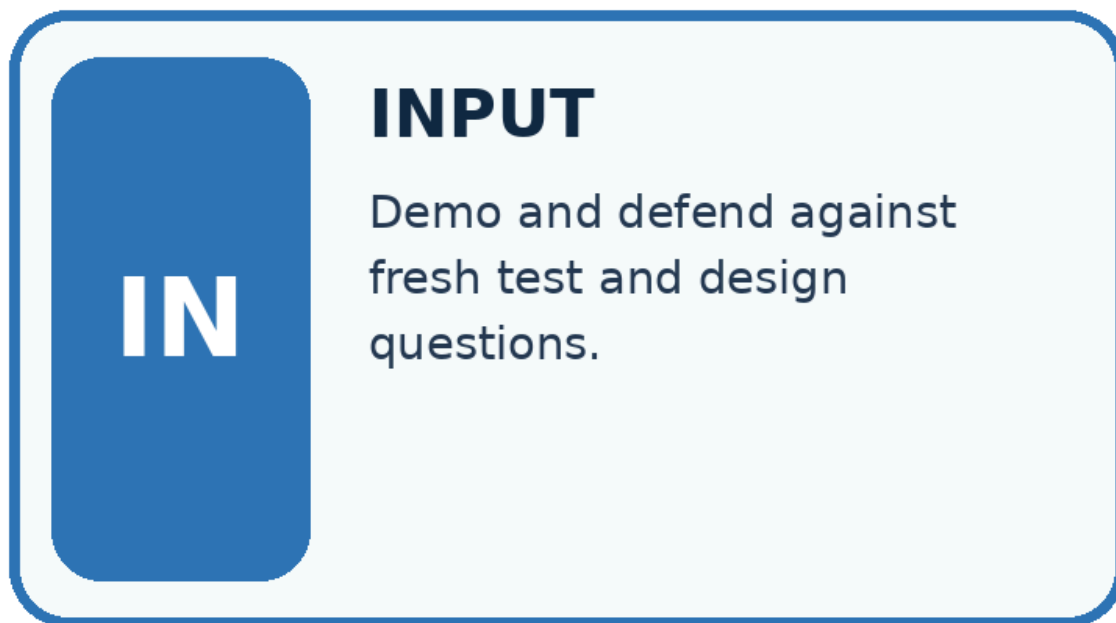
Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



SYSTEM

A creator explains architecture, evidence, limitations, provenance, and responsible use.



OUT

EVIDENCE

Demo and defend against fresh test and design questions.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Link each claim to a recorded result. State the distinction between retrieval relevance, claim support, fallback and permissions. Give a reproduction packet and answer challenges without inventing missing measurements.

Generative systems defense — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: System reports 9/10 retrieved relevant records, 7/10 supported answers and zero actions in a read-only run.

Check or calculation: Link each claim to a recorded result. State the distinction between retrieval relevance, claim support, fallback and permissions. Give a reproduction packet and answer challenges without inventing missing measurements.

Conclusion and limit: Retrieval rate 90% and support rate 70% differ. Action trace supports zero actions in that run; no timing evidence supports speed. Correct the full-correctness and speed claims.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected learner request



AMBIGUOUS
Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Only the retrieval rate is put on the presentation poster.

Prediction: The poster omits the lower supported-answer rate and cannot imply 80% fully correct answers.

Reason: Retrieving useful evidence is an intermediate step, not proof that the final answer uses it correctly.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

ARCHITECTURE

components and
information flow

3

LIMITATION

condition where
performance weakens

4

TRANSFER TEST

fresh case not rehearsed

Word · My own example · Not this / boundary

Word: defense

My own example: An explanation of a project decision supported by evidence and an account of its limitations.

Not this / boundary: _____

Word: architecture

My own example: The arrangement of a system's parts and the connections between them.

Not this / boundary: _____

Word: limitation

My own example: A condition or missing capability that restricts what a result can establish or a system can safely do.

Not this / boundary: _____

Word: transfer test

My own example: A fresh task that checks whether learning can be applied without the original supports.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Defense claim

Meaning: A statement tied to inspectable project evidence

Word: Reproduction packet

Meaning: Artifacts needed to repeat a reported procedure

Word: Unmeasured outcome

Meaning: A property for which the experiment supplies no observation

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 36 • WORKED EVIDENCE

- 1 Trace one question through retrieval, generation, validation, and citation.
- 2 Recompute one evaluation score and show source support.
- 3 Run an unseen missing-source or conflicting-source case.
- 4 State intended use, do-not-use case, privacy/provenance controls, and human responsibility.

1. Trace one question through retrieval, generation, validation, and citation.
2. Recompute one evaluation score and show source support.
3. Run an unseen missing-source or conflicting-source case.

4. State intended use, do-not-use case, privacy/provenance controls, and human responsibility.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

System reports 9/10 retrieved relevant records, 7/10 supported answers and zero actions in a read-only run. Presenter claims 90% fully correct and faster than baseline.

Retrieval rate 90% and support rate 70% differ. Action trace supports zero actions in that run; no timing evidence supports speed. Correct the full-correctness and speed claims.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Creator Defense

Defend a study-buddy project using separate claims and matching evidence.

The facts we will use

Set A: 15 answerable questions, with 12 relevant retrievals and 9 supported answers. Separate set B: 5 missing-source questions, with 4 appropriate fallbacks. No speed measurements exist.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Creator Defense

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Retrieval claim

Relevant retrieval rate is $12/15=80\%$.



2. Answer claim

Supported-answer rate is $9/15=60\%$; retrieval success is not answer success.



3. Fallback claim

Appropriate fallback is $4/5=80\%$. Make no measured speed claim.

CHANGE ONE THING

Only the retrieval rate is put on the presentation poster.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Retrieval claim

What happens and why: Relevant retrieval rate is $12/15=80\%$.

Step: 2. Answer claim

What happens and why: Supported-answer rate is $9/15=60\%$; retrieval success is not answer success.

Step: 3. Fallback claim

What happens and why: Appropriate fallback is $4/5=80\%$. Make no measured speed claim.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Appropriate fallback is $4/5=80\%$. Make no measured speed claim. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

These invented counts illustrate defense structure; the class must replace them with its own documented evidence.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: A smooth demo proves the system is reliable. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

A defense should preserve negative evidence and uncertainty rather than treating polished presentation as proof.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Generative systems defense”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Generative systems defense. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which numbers would you display to make the defense honest? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN

Present architecture and evidence bundle, accept a fresh case, explain a failure without hiding it, and defend one tradeoff.

Present architecture and evidence bundle, accept a fresh case, explain a failure without hiding it, and defend one tradeoff.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY

1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY

Final product card documents purpose, system map, sources, evals, limits, risks, and human oversight.

THIS WEEK’S CHECKPOINT Final product card documents purpose, system map, sources, evals, limits, risks, and human oversight.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

Demo and defend against fresh test and design questions.

CHALLENGE Demo and defend against fresh test and design questions.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 36 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Ten successful demos lead to a never hallucinates claim.

The question

Write a defensible conclusion.

Worked explanation

Report those ten specified passes, show evidence and fallback, and state untested conditions. Demonstrate an unanswerable question.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Defend a bounded claim

Evidence or stage · Value or result

Evidence or stage: Result

Value or result: Ten specified demonstrations passed

Evidence or stage: Evidence

Value or result: Show sources and checks

Evidence or stage: Limit

Value or result: Other conditions remain untested

Evidence or stage: Demonstration

Value or result: Show an unanswerable question

Ten successful demos lead to a never hallucinates claim.

Write a defensible conclusion.

Report those ten specified passes, show evidence and fallback, and state untested conditions.

Demonstrate an unanswerable question.

14. Try a new case independently

Independent case: Apply a new case independently

A defense has retrieval 8/10, support 6/10, fallback 3/4 and no measured latency. State the three rates and reject one unsupported claim.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 36 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

An audience asks whether your assistant is always correct. Give an evidence-based defense using results and a limit.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Final product card documents purpose, system map, sources, evals, limits, risks, and human oversight.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Your assistant passes 9/12 final questions, including all six direct facts but only 3/6 ambiguous or absent-evidence cases. Report results and argue for a focused next step.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.